

Abstract No.

IL-12-Armed EGFRvIII-Retargeted Oncolytic HSV Promotes Agnostic Vaccination in Heterogeneous Murine Glioblastoma

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Tumor heterogeneity represents one of the major obstacles to targeted glioblastoma therapies, since any treatment directed against a specific molecular target may spare tumor cell populations lacking its expression. We hypothesized that this limitation could be overcome by using a retargeted oncolytic HSV (ReHV) to deliver IL-12 selectively within the tumor. While viral infection and oncolysis remain restricted to target-positive cells, local IL-12-mediated immune stimulation is expected to induce an agnostic vaccination, extending the antitumor response to cells that cannot be directly infected by the virus.

We tested an EGFRvIII-retargeted ReHV expressing murine IL-12 in immunocompetent mice orthotopically transplanted with glioblastoma cells. Two independent models entirely composed of EGFRvIII-positive cells were analyzed together with a heterogeneous model containing both EGFRvIII-positive and EGFRvIII-negative cells. Animals were treated seven days after transplantation and tumor progression was monitored weekly by MRI.

In both homogeneous models, treatment consistently delayed tumor growth and substantially prolonged survival, increasing median survival by approximately 1.5- to 2-fold compared with untreated controls, with complete tumor eradication observed in a subset of animals. Importantly, therapeutic efficacy was also maintained in heterogeneous tumors, despite the presence of cells lacking the viral target. In this setting, median survival more than doubled, and almost half of the treated animals showed long-term survival without detectable tumor. MRI analysis confirmed a marked delay in tumor growth following treatment. Preliminary rechallenge experiments in long-surviving mice also showed protection against subsequent tumor transplantation.

These findings support the ability of IL-12-armed ReHV to convert selective viral targeting into a broader immune-mediated response, providing experimental evidence for an agnostic vaccination strategy capable of overcoming glioblastoma heterogeneity.



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